

Q.No:6

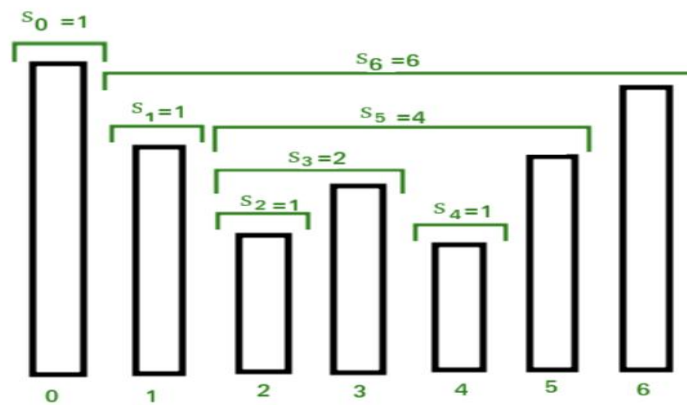
Question description

The **stock span problem** is a financial problem where we have a series of n daily price quotes for a stock and we need to calculate span of stock's price for all n days.

The span S_i of the stock's price on a given day i is defined as the maximum number of consecutive days just before the given day, for which the price of the stock on the current day is less than or equal to its price on the given day.

For example, if an array of 7 days prices is given as {100, 80, 60, 70, 60, 75, 85},

then the span values for corresponding 7 days are {1, 1, 1, 2, 1, 4, 6}



Constraints

$0 < n < 1000$

$0 < \text{price}[i] < 1000000$

Input Format:

First line indicates the number of days

second line indicates the price quoted for above mentioned days

Output Format:

▼ Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 6 9 8 7 6 5 4	INPUT (STDIN) 6 10 41 15 90 10 70
EXPECTED OUTPUT 1 1 1 1 1 1	EXPECTED OUTPUT 1 2 1 4 1 2

▼ Mandatory Test Cases

Test Case 1	Test Case 2
KEYWORD void printArray(int arr[],int n)	KEYWORD void calculateSpan(int price[],int n,int S[])

▼ Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY 4	TOKEN COUNT 220	NLOC 35

Code:

```
#include <stdio.h>

#define MAX 1000

void calculateSpan(int price[], int n, int S[])
{
    int stack[MAX];
    int top = -1;
    S[0] = 1;
    stack[++top] = 0;

    for (int i = 1; i < n; i++)
    {
        while (top != -1 && price[stack[top]] <= price[i])
        {
```

```

top--;
}

if (top == -1)
{
S[i] = i + 1;
}
else
{
S[i] = i - stack[top];
}
stack[++top] = i;
}
}

void printArray(int arr[], int n)
{
for (int i = 0; i < n; i++)
{
printf("%d ", arr[i]);
}
}

int main()
{
int n;
int price[MAX];
int S[MAX];

scanf("%d", &n);

for (int i = 0; i < n; i++)
{
scanf("%d", &price[i]);
}

calculateSpan(price, n, S);

printArray(S, n);

return 0;
}

```

Output:

Test_case 1:

```
6
9 8 7 6 5 4
1 1 1 1 1 1 %
```

Test_case 2;

```
6
10 41 15 90 10 70
1 2 1 4 1 2 %
```